

GsoC 2026 Organization: JSON Schema

Automated Observability For The JSON Schema Ecosystem

GitHub: <https://github.com/MakikoOhashi>

Qualification Task Repository:

<https://github.com/MakikoOhashi/json-schema-ecosystem-metrics>

Project Summary

This proposal aims to help establish a starting point for observability in the JSON Schema ecosystem. In the qualification task, I built one primary metric that is easy to refresh over time, and I also experimented with a repository-level comparison metric to explore what kinds of signals may be useful for understanding ecosystem change and adoption. I also reviewed the existing proof-of-concept in `projects/initial-data/` and tried to identify which ideas seem worth keeping and which parts may benefit from a clearer redesign.

Problem

The JSON Schema ecosystem is already widely used, and I think there is real value in making its changes easier to follow over time. Signals showing which implementations are being used, where activity appears to be growing, and where more support may be helpful could be useful for the community if they can be collected in a repeatable way. At the same time, it is not straightforward to define the ecosystem boundary itself, and the meaning of any adoption metric depends heavily on what is chosen as the denominator and the numerator. For that reason, I think it makes sense to begin with small, repeatable observations rather than trying to define a perfect measurement from the start.

Proposed Work

In this proposal, I would like to help build a lightweight starting point for observing the JSON Schema ecosystem. In the qualification task, I approached observability in two directions. The vertical direction focused on change over time, using one important validator signal as a primary metric. The horizontal direction explored how visible explicit JSON Schema usage might be across repository cohorts. The result was that the time-based signal was relatively stable, while the repository-comparison work was much more sensitive to cohort definition and GitHub API rate limits. Based on that experience, I would approach the project by stabilizing repeatable primary signals first, while gradually improving the signal design needed for a more meaningful adoption ratio over time.

Qualification Task

Part 1: Proof of Concept

<https://github.com/MakikoOhashi/json-schema-ecosystem-metrics>

For Part 1, I used `ajv` npm downloads as the primary metric. This does not directly measure the whole JSON Schema ecosystem, but it provides a practical proxy for validator-level adoption activity within a JavaScript/TypeScript-facing slice of the ecosystem. I fetched the data from the npm downloads API, saved it as structured JSON, and visualized it in a dashboard using a 7-day rolling average so the recent trend would be easier to read. I found this to be a useful way to represent the “time/change” side of observability in a compact form.

I also explored a second, more experimental metric at the repository level. For that work, I compared broad and focused repository cohorts and tested signals such as `\*.schema.json`, `"\$schema"`, and dependency markers to see how visible explicit JSON Schema usage might be. What I learned is that no single marker seems strong enough on its own, and that the meaning of the result changes significantly depending on how the cohort is defined. I also ran into GitHub API rate limits, which made the horizontal metric much more sensitive operationally than the primary time-series metric. Because of that, I treat this exploratory work less as a finished adoption metric and more as a signal-design exercise toward something closer to a future `JSON-bearing repositories / JSON-Schema-positive repositories` ratio.

Part 2: Existing Code Evaluation

For Part 2, I reviewed the existing proof-of-concept in `projects/initial-data/`. After looking through it and trying to run it, my impression was that the overall idea of collecting and recording ecosystem candidates is a helpful starting point. At the same time, the current topic-based discovery and CSV-oriented output seem likely to introduce noise and make automation harder than necessary if they are used as a long-term observability foundation. My current view is that the repository-discovery mindset and exploratory spirit are worth keeping, while the data model, metric definitions, and output structure would likely benefit from a clearer and more structured redesign.

Deliverables

The qualification-task deliverables are Node.js scripts, structured JSON outputs, a single-page dashboard, a README, Part 1 notes, and a Part 2 evaluation. For the project itself, I would hope to build repeatable data-collection steps, outputs that can be refreshed on a regular schedule, and a lightweight reporting layer that makes the meaning and limitations of each signal easy to understand. One goal would be to make it clearer which metrics describe time/change and which ones are better suited to relative-position or visibility questions.

Timeline

In the first phase, I would focus on clarifying the current prototype, refining metric definitions, and stabilizing a few repeatable primary signals. In the middle phase, I would work on improving exploratory metrics, especially around cohort definition and signal design, and make the collection flow easier to refresh on a weekly basis. In the later phase, I would refine the reporting layer and documentation so that the work can serve as a more usable base for broader ecosystem observability in the future. My general approach would be to stabilize a narrower slice first and only then expand coverage more carefully.

Risks

The main difficulty, as I currently understand it, is defining the ecosystem boundary and the meaning of the metrics. JSON Schema adoption is not easy to represent through a single repo count or a single package download series, because the interpretation depends heavily on what is counted as the broader JSON-using world and what counts as a strong schema signal. There are also operational constraints such as GitHub API rate limits, which make reproducibility and observability harder to balance. Because of that, I would prefer an incremental approach that makes signal meaning and limitations explicit rather than aiming for perfect coverage immediately.

AI Assistance

During the qualification task, I used AI mainly for thinking support, comparing metric ideas, drafting parts of the README and explanatory text, and exploring implementation options. However, I have tried to make sure that I can explain the purpose, limitations, and tradeoffs of the code and metrics in my own words. In particular, for the exploratory metrics, I did not simply accept suggested directions as-is; I changed direction based on actual outputs, actual API behavior, and my own understanding of what the metrics were or were not really showing.